COURSE INFORMATION

School/Faculty:	Computing/Engineering	**Page:**	1 of 7
Program name:	Bachelor in Software Engineering with Honors		
Course code:	SCST1223	**Academic Session/Semester:**	2025/26/2
Course name:	Probability and Statistical Data Analysis	**Pre/co requisite (course name and code, if applicable):**	
Credit hours:	3		

Course synopsis	This course is designed to introduce some statistical techniques as tools to analyse the data. In the beginning the students will be exposed with various forms of data. The data represented by the different types of variables are derived from different sources; daily and industrial activities. The analysis begins with the data representation visually. The course will also explore some methods of parameter estimation from different distributions. Further data analysis is conducted by introducing the hypothesis testing. Some models are employed to fit groups of data. At the end of course the students should be able to apply some statistical models in analysing data using available software.
Course coordinator (if applicable)	Dr Seah Choon Sen

Course lecturer(s)		**Name**	**Office**	**Contact no.**	**E-mail**
01	SCSEH	Dr. Shamini A/p Raja Kumaran	403-22 (N28)	6012-905 0323	shamini.rk@utm.my
02		Dr. Seah Choon Sen	02-13-01 (N28a)	6016-414 5774	seahcs@utm.my
03					
04		Dr. Nur Eiliyah @ Wong Yee Leng	02-02-01 (N28a)	6013-521 3941	nureiliyah@utm.my

Mapping of the Course Learning Outcomes (CLO) to the Programme Learning Outcomes (PLO), Teaching & Learning (T&L) methods and Assessment methods:

No.	**CLO**	**PLO (Code)**	***Taxonomies and **generic skills**	**T&L methods**	****Assessment methods**
CLO1	Analyse the relationship of different variables using descriptive statistics	PLO1 (KW)	C4 (SC1)	Lecture, Project-based learning	Q, ASG, PR, T
CLO2	Analyze data sets using different statistical methods	PLO7 (NS)	C4 A(IT3)	Lecture, Project-based learning	Q, ASG, FE, PR
CLO3	Propose potential improvement based on the statistical analysis	PLO9 (PRS)	A3 A(ES2)	Project-based learning	PR
***T – Test; Q – Quiz; ASG – Assignment; PR – Project; FE – Final Exam					

Prepared by: Name: Dr. Suhaila Mohamad Yusuf Signature: Date:	Certified by: Name: Signature: Date:

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Details on Innovative T&L practices:

No.	Type	Implementation
1.	Active learning	Conducted through in-class activities
2.	Project-based learning	Conducted through project assignment. Tasks are given in sequential steps throughout the semester. Students in a group of 3/4 are given opportunity to collect data and perform some analysis and present it in suitable manner. The report must be given in the form of written report or short paper.

Weekly Schedule:

WEEK / DATE	TOPICS	ACTIVITIES
WEEK 1 16 – 20 Mar 25 **Hari Raya AidilFitri (21/3 & 22/3 – Saturday, Sunday)* *Online*	**Chapter 1: Introduction to Statistics** **1.1: Introduction** 1.1.1 Descriptive and Inferential Statistics. 1.1.2 Population and Sample.	
WEEK 2 23 – 27 Mar 25 ** Sultan of Johor's Birthday (23/3 - Monday)* *Online*	**1.2: Data** 1.2.1 Data Analysis Process. 1.2.2 Data Sources (Primary and Secondary data). 1.2.3 Types of Data (Qualitative, Quantitative, Discrete and Continuous data). 1.2.4 Data Scale and Measurement (Nominal, Ordinal, Interval, Ratio).	**GROUP PROJECT 1 BRIEFING**
WEEK 3 30 Mar – 03 Apr 25 *Physical*	**Chapter 2: Data Description** **2.1: Presenting Qualitative Data** 2.1.1 Frequency Distributions, Bar and Pie Charts. **2.2: Presenting Quantitative Data** 2.2.1 Frequency Distributions, Histograms, Stem-and-Leaf, Box Plot.	One hour of exercise/tutorial each week, starting from week 3.
WEEK 4 06 – 10 Apr 25	**Chapter 3: Descriptive Statistics** **3.1: Measurement of Central Tendency** 3.1.1Mean, Median, Mode, Quartile and Percentile. **3.2: Measurement of Dispersion** 3.2.1 Range, Variance, Standard Deviation.	**QUIZ 1** (Chapter 3) (5%)

Physical	3.2.2 Skewness and Kurtosis.	
WEEK 5 13 – 17 Apr 25 *Physical*	**Chapter 4: Probability, Random Variables and Probability Distributions** **4.1: Probability** 4.1.1 Overview of Probability. **4.2: Random Variables and Probability Distributions** 4.2.1 Discrete and Continuous Random Variables. 4.2.2 Discrete and Continuous Variables Probability Distribution. 4.2.3 Binomial, Geometric and Poisson Distributions. 4.2.4 Normal Distribution.	**Assessment:** **ASSIGNMENT 1** (Chapter 1-4) (5%)
WEEK 6 20 – 24 Apr 25 *Physical*	**Lab Session: Introduction to Statistical Tools (R Programming)** **Topic 1: Introduction** **Topic 2: Basic Analysis** i. Frequencies Analysis. ii. Descriptive Analysis.	**Mid-Term Test:** Date: 21 April 2025 Day: Tuesday Venue: P19 Time: 8:00 pm -10:00 pm (15%)
WEEK 7 27 Apr – 01 May 25 **Labour Day (1/5 - Friday)* *Physical*	**Chapter 5: Hypothesis Testing** **5.1: Point Estimation** 5.1.1 Point Estimator 5.1.2 Interval Estimator	**SUBMISSION:** **PROJECT 1** **(5%)**
WEEK 8 04 – 08 May 25	**MID-SEMESTER BREAK**	
WEEK 9 11 – 15 May 25 *Physical*	**5.2: Hypothesis Testing for 1 Sample** 5.2.1 Hypothesis Statement and Decision Rule 5.2.2 Errors of Decision 5.2.3 Hypothesis Testing	
WEEK 10 18 – 22 May 25 *Physical*		
WEEK 11 25 – 29 May 25 **Hari Raya Haji (27/5 - Wednesday)* *Physical*	**5.3: Hypothesis Testing for 2 Samples** 5.3.1 Hypothesis Statement 5.3.2 Hypothesis Testing	

WEEK 12 01 – 05 June 25 **Wesak Day (31/5 - Sunday)* *Agong's Birthday (2/6-Monday)* *Online*		**QUIZ 2** (Chapter 5.1 – 5.2) (5%)
WEEK 13 08 – 12 June 25 *Physical*	**Chapter 6: Chi-Square Test and Contingency Analysis** **6.1: Multinomial Experiment and Goodness-of-Fit Test** 6.1.1 Multinomial Experiment 6.1.2 Goodness-of-Fit Test **6.2: One-way Contingency Table** 6.2.1 Categories with equal frequencies/probabilities 6.2.2 Categories with unequal frequencies/probabilities **6.3: Two-way Contingency Table** 6.3.1 Chi-Square Test of Independence	
WEEK 14 15 – 19 June 25 **Awal Muharram (17/6 – Wednesday)* *Physical*	**Chapter 7: Correlation and Regression** **7.1: Correlation** 7.1.1 Correlation Analysis. 7.1.2 Pearson's Correlation. 7.1.3 Spearman's Correlation. **7.2: Regression** 7.2.1 Types of Regression Models. 7.2.2 Population Linear Regression. 7.2.3 The Least Square Equation. 7.2.4 Coefficient of Determination. 7.2.5 Standard Error and Standard Deviation.	**ASSIGNMENT 3** (Chapter 7 & 8) (5%) **SUBMISSION: PROJECT 2 (10%)**
WEEK 15 22 – 26 June 25 *Physical*	**Chapter 8: Analysis of Variance (ANOVA)** **One-way ANOVA** 8.1 ANOVA with Equal Sample Sizes. 8.2 ANOVA with Unequal Sample Sizes	**PROJECT 2 PRESENTATION**
WEEK 16 29 June – 03 July 2025	**STUDY WEEK**	
WEEK 17 – 19 06 July – 26 July 25	**FINAL EXAM WEEK**	

Transferable skills (generic skills learned in course of study which can be useful and utilised in other settings):

Team working
Oral & Written communication

Student learning time (SLT) details:

Distribution of student Learning Time (SLT) Course content outline	Guided Learning (Face to Face)				Teaching and Learning Activities: Guided Learning Non-Face to Face	Teaching and Learning Activities: Independent Learning Non-Face to face	TOTAL SLT
CLO	**L**	**T**	**P**	**O**			
CLO 1	15	2	5		10	8	**40h**
CLO 2	13	3	2		10	20	**48h**
CLO3				2		17	**19h**
Total SLT	**28**	**5**	7	**2**	**20**	**45**	**107**

Continuous Assessment		PLO	Percentage	Total SLT
1	ASG 1	CLO1 / PLO1 (KW)	5	2h
2	ASG 2	CLO2 / PLO7 (NS)	10	2h
3	ASG 3	CLO 2 / PLO7 (NS)	5	1h
4	Quiz 1	CLO1 / PLO1 (KW)	5	**1h**
5	Quiz 2	CLO2 / PLO7 (NS)	5	
5	Test	CLO 1 / PLO1 (KW)	15	**2h**
6	Project 1	CLO 1 / PLO1 (NS) CLO 3 / PLO9 (PRS)	3 2	1h As in CLO3 (1h)
7	Project 2	CLO 2 / PLO7 (NS) CLO 3 / PLO9 (PRS)	7 3	1h As in CLO3 (1h)
Final Assessment			Percentage	Total SLT
1	Final Exam	CLO2 / PLO7 (NS)	40	**3h**
Grand Total			**100**	**120h**

L: Lecture, T: Tutorial, P: Practical, O: Others

Special requirement to deliver the course (e.g: software, nursery, computer lab, simulation room):

Computer lab with *R* software

Learning resources:

Text book (if applicable)

Main references

1. Roxy Peck, Chris Olsen, Jay Devore, Introduction to Statistics and Data Analysis, 6th Edition, Brooks/Cole Cengage Learning, 2020.
2. Mario F. Triola, Elementary Statistics, 12th Edition, Pearson, 2012.
3. Neil A. Weiss, Elementary Statistics, 8th Edition, Pearson, 2011.

Additional references

Any suitable Statistics website and books.

Online

http://elearning.utm.my

Academic honesty and plagiarism: *(Below is just a sample)*

Assignments are individual tasks and NOT group activities (UNLESS EXPLICITLY INDICATED AS GROUP ACTIVITIES) Copying of work (texts, simulation results etc.) from other students/groups or from other sources is not allowed. Brief quotations are allowed and then only if indicated as such. Existing texts should be reformulated with your own words used to explain what you have read. It is not acceptable to retype existing texts and just acknowledge the source as a reference. Be warned: students who submit copied work will obtain a mark of **zero** for the assignment and disciplinary steps may be taken by the Faculty. It is also unacceptable to do somebody else's work, to lend your work to them or to make your work available to them to copy.

Other additional information (Course policy, any specific instruction etc.):

1. Attendance is compulsory and will be taken in every lecture session. Student with less than 80% of total attendance is not allowed to sit for final exam.
2. Students are required to behave and follow the University's dressing regulation and etiquette all the time.
3. Exercises and tutorial will be given in class and some may be taken for assessment. Students who do not do the exercise will lose the coursework marks for the exercise.
4. Assignments must be submitted on the due dates. Some points will be deducted for late submissions. Assignments submitted three days after the due date will not be accepted.
5. Make up exam will not be given, except to students who are sick and submit medical certificate confirmed by UTM panel doctors. Make up exam can only be given within one week of the initial date of exam.
6. To pass this course, students must pass both Continuous and Final assessment.

Disclaimer:

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REVIEW OF L&T ACTIVITIES TO INCLUDE ONLINE LEARNING

Course learning outcome	Guided Learning FTF hours (from CI)	Guided Learning FTF hours completed	Online Learning hours			
			Activities	**Type of time spent**	**Estimated time**	**Total time**
CLO1 Analyse the relationship of different variables using descriptive statistics	3hrs x 3weeks (9)	0	9			
			Online lecture through Webex Meeting platform (Chapter 1 & 2)	The time spent in synchronous live interaction	2hrs (Chap 1.1) 2hrs (Chap 1.2) 2hrs (Chap 2)	6hrs
			Live interaction with students to brief assignment instruction and give feedback on assignment progress	The time spent in synchronous live interaction	1hr (Project 1)	1hr
			Live Interaction with students to discuss students' work	Time spent on instructional activities	30mins (Chap 1.1) 30mins (Chap 1.2) 1hr (Chap 2)	2hrs